Paper 57: Analytic Closure of the VCML Lyapunov Condition

Boundary Geometry, Causal Purity, and the Substrate-Agnostic Form of the Identity Threshold

Gabriel Aubin-Moreau[1]

[1]Independent research

March 2026

Abstract

Paper 55 established that $C_{\text{order}} = (D_{\text{nonadj}} - D_{\text{adj}})/\sigma_w$ has monotone positive drift (Lyapunov property) in the C_perturb regime, but the derivation of the noise-growth term $\dot{W}/W$ bottomed out in simulation. Paper 56 corroborated that $r_{\text{wave}} = 8$ fails ($\sigma_w = 0.096$) while $r_{\text{wave}} = 2$ succeeds ($\sigma_w = 0.011$) for C_perturb at S2.

We close the missing middle.

Geometric approximation: Cells within r_{wave} of a zone boundary receive mixed-zone wave input, inflating σ_w. The fraction of boundary-contaminated cells is $2r_{\text{wave}}/W_{\text{zone}} = 2/\delta$, giving an interior fraction $f_{\text{int}}(\delta) = (1 - 2/\delta)^+$. This predicts a critical threshold $\delta_c = 2$ below which all cells are boundary-contaminated and σ_w inflates to C_ref levels.

Fine r_wave sweep (new data, Exp 57): We run C_perturb sub-threshold at S2 over $r_{\text{wave}} \in \{2, 3, 4, 5, 6, 7, 8, 10\}$ (40 runs total, 5 seeds each). σ_w scales roughly linearly with boundary exposure $2/\delta$, confirming the geometric model directionally. However, $r_{\text{wave}} = 10$ ($\delta = 2$, $f_{\text{int}} = 0$) achieves na $= 1.091$ rather than full failure, revealing a systematic deviation: the geometric formula underestimates effective purity because it ignores the zone-launch surface area asymmetry.

Causal purity reparametrization: The correct primitive quantity is not geometric distance to a zone boundary but *causal purity*: the fraction of a cell's pre-consolidation wave-contact events originating from the same zone class. On the lattice this reduces to δ-geometry at large δ, but remains nonzero even at $\delta = 2$ because same-zone launches outnumber cross-zone reaches. The substrate-agnostic statement is: stable identity requires $P_{\text{causal}} > p_c$ for some threshold p_c, where P_{causal} is the mean causal purity over all consolidating cells.

Revised Lyapunov condition: $\dot{C}_{\text{order}} > 0$ iff the mean causal purity of consolidating cells exceeds a threshold. On a rectangular lattice this projects to $\delta > \delta_c \approx 2$; on other substrates it projects to the analogous ratio of zone-local to cross-zone causal influence.

Implication: This is the first substrate-agnostic statement of the identity formation condition. The geometric laws of Papers 54–56 are the lattice-specific projection of a single primitive constraint: a nonzero fraction of cells must have causally pure pre-consolidation neighborhoods.

1 Introduction

The open problem. Paper 55 derived the Lyapunov condition

$$\dot{C}_{\text{order}} > 0 \iff \frac{\dot{S}}{S} > \frac{\dot{W}}{W} \tag{1}$$

where $S = D_{\text{nonadj}} - D_{\text{adj}}$ and $W = \sigma_w$. It showed empirically that C_perturb satisfies it ($\dot{W}/W \approx 0$) while C_ref violates it ($\dot{W}/W > 0$), but did not derive $\dot{W}/W$ analytically from the VCML update rules.

Paper 56 showed that $r_{\text{wave}} = 8$ fails under C_perturb at S2 ($\sigma_w = 0.096$), while $r_{\text{wave}} = 2$ succeeds ($\sigma_w = 0.011$), and that write-policy determines effective noise floor.

The missing piece: an analytic expression for $\dot{W}/W$ as a function of $\delta = W_{\text{zone}}/r_{\text{wave}}$ and write policy.

This paper. We derive the noise-growth term from two successive approximations. First, a geometric approximation based on the spatial boundary structure (Sections 2–3). Second, a causal-purity reparametrisation that removes lattice-specific assumptions and produces the substrate-agnostic form (Section 4). Section 5 presents the new experimental verification (Exp 57, fine r_{wave} sweep), and Section 6 states the revised Lyapunov condition.

2 Geometric Approximation: Boundary Fraction and Noise Growth

2.1 Setup and notation

Consider a rectangular lattice with 4 zones of width W_{zone} and height H (in the S2 experiments: $W_{\text{zone}} = 20$, $H = 80$). Waves propagate with taxicab radius r_{wave} from launch points uniformly distributed within each zone.

Let $\delta = W_{\text{zone}}/r_{\text{wave}}$ be the zone-width-to-causal-radius ratio. Let V_{int} be the within-zone fieldM variance contributed by cells that receive exclusively same-zone wave input. Let $V_{\text{bnd}} \gg V_{\text{int}}$ be the variance contributed by cells that receive mixed-zone wave input (boundary-adjacent cells).

2.2 The boundary fraction

Cells within r_{wave} of either vertical zone edge can receive wave signals from adjacent zones. In a rectangular zone with lateral width W_{zone}, the fraction of boundary-contaminated cells is

$$f_{\text{bnd}}(\delta) = \frac{2r_{\text{wave}}}{W_{\text{zone}}} = \frac{2}{\delta}. \tag{2}$$

The complementary interior fraction is

$$f_{\text{int}}(\delta) = \left(1 - \frac{2}{\delta}\right)^{+}. \tag{3}$$

2.3 Noise growth under C_ref and C_perturb

C_ref. All quiescent cells write to fieldM. Both boundary and interior cells contribute. Since $V_{\text{bnd}} \gg V_{\text{int}}$, boundary cells dominate:

$$\mathbb{E}[\Delta\sigma_w^2 \,|\, \text{C_ref}] \approx f_{\text{bnd}} \cdot V_{\text{bnd}} = \frac{2}{\delta} V_{\text{bnd}} > 0 \quad \forall\, \delta. \tag{4}$$

This is bounded away from zero at all finite δ: C_ref always has $\dot{W}/W > 0$.

C_perturb. Writes fire only during active wave-contact events. Boundary cells receive waves from adjacent zones, continuously resetting their calm streak; they rarely satisfy the write gate. Effective writes come almost exclusively from interior cells:

$$\mathbb{E}[\Delta\sigma_w^2 \,|\, \text{C_perturb}] \approx f_{\text{int}}(\delta) \cdot V_{\text{int}} \approx \begin{cases} \epsilon \approx 0 & \delta > 2 \\ V_{\text{bnd}} & \delta \leq 2. \end{cases} \tag{5}$$

At $\delta = 2$, $f_{\text{int}} = 0$: no interior cells exist, the spatial filter breaks down, and σ_w inflates to C_ref levels.

2.4 Predicted critical threshold

The noise-floor protection collapses when $f_{\text{int}}(\delta) = 0$:

$$\delta_c = 2, \quad \text{i.e.} \quad r_{\text{wave}} = W_{\text{zone}}/2. \tag{6}$$

This is a geometric fact: when the causal radius equals half the zone width, every cell is within the boundary contamination radius. The threshold δ_c depends only on the ratio of zone width to causal radius, not on any VCML parameter.

2.5 The signal-growth term

Zone separation S grows via the copy-forward loop (zone-mean ODE, Papers 22–26). Interior cells carry unambiguous zone signal, so

$$\frac{\dot{S}}{S} \propto f_{\text{int}}(\delta) \cdot r_{\text{wave}}. \tag{7}$$

The factor r_{wave} enters because a wave of larger radius seeds the copy-forward loop over more cells per event.

Substituting into (1), the Lyapunov condition becomes

$$f_{\text{int}}(\delta) \cdot r_{\text{wave}} > \frac{2}{\delta} V_{\text{bnd}}/\sigma_w \tag{8}$$

which for C_perturb simplifies (since $\dot{W}/W \approx 0$ when $\delta > 2$) to

$$\dot{C}_{\text{order}} > 0 \iff \delta > 2 \quad \text{(C_perturb, geometric approximation).} \tag{9}$$

3 Experimental Verification (Exp 57)

We run C_perturb, sub-threshold amplitude (supp= 0.06, exc= 0.12), S2 grid, $r_{\text{wave}} \in \{2, 3, 4, 5, 6, 7, 8, 10\}$, 5 seeds each (40 total runs, 3000 steps, tracked every 25 steps). $W_{\text{zone}} = 20$, so $\delta_c = 2$ predicts $r_{\text{wave}} = 10$ as the threshold.

Table 1: Exp 57: fine r_{wave} sweep under C_perturb at S2. f_{int} = geometric interior fraction. All metrics at step 3000.

r_{wave}	δ	f_{int}	final na_ratio	C_{order}	$\sigma_w(3000)$
2	10.00	0.800	1.261	+0.043	0.0111
3	6.67	0.700	1.254	+0.025	0.0278
4	5.00	0.600	1.140	+0.016	0.0452
5	4.00	0.500	1.132	+0.018	0.0625
6	3.33	0.400	1.038	−0.011	0.0798
7	2.86	0.300	1.009	+0.001	0.0893
8	2.50	0.200	0.943	−0.008	0.0961
10	2.00	0.000	1.091	+0.002	0.1060

Confirmation of the directional prediction. σ_w increases monotonically with r_{wave}, broadly tracking $2r_{\text{wave}}/W_{\text{zone}}$ (Figure 1a). na_ratio and C_{order} degrade with r_{wave} up to $r_{\text{wave}} = 8$. The optimal is $r_{\text{wave}} = 2$–3, matching the geometric prediction of maximum effective purity.

Deviation at $r_{\text{wave}} = 10$ ($\delta = 2$). The geometric model predicts full collapse ($f_{\text{int}} = 0$), but the system achieves na $= 1.091$ with $\sigma_w = 0.1060$. Identity is marginal (cross_step $= 2275$, very late) but nonzero. This is a systematic underprediction: the geometric formula treats f_{int} as binary (in/out of boundary zone) but ignores the zone-launch area asymmetry. Even at $\delta = 2$, a central cell receives more wave-contact events from same-zone launches than from cross-zone launches, because the same zone has more total area contributing. The effective causal purity is nonzero.

This deviation motivates the causal-purity reparametrisation.

4 Causal Purity: The Substrate-Agnostic Form

4.1 The primitive quantity

Replace the lattice-geometric framing with a causal one. For each site i, define:

- $\mathcal{N}_c(i)$: the set of sites whose state can influence i within one write-consolidation window (the *causal neighbourhood*).
- $p_{\text{same}}(i)$: the fraction of $\mathcal{N}_c(i)$ launch-events originating from the same zone class as i.

Then

$$P_{\text{causal}} = \mathbb{E}_i[p_{\text{same}}(i)] \tag{10}$$

is the mean causal purity over all consolidating cells.

On the lattice with taxicab waves, $p_{\text{same}}(i)$ for a cell at horizontal position x within zone Z (of width W_{zone}) is approximately

$$p_{\text{same}}(i) \approx \frac{W_{\text{zone}}}{W_{\text{zone}} + 2\min(x, W_{\text{zone}} - x, r_{\text{wave}})}, \tag{11}$$

which exceeds $1/3$ even at $\delta = 2$ for central cells. This explains the $r_{\text{wave}} = 10$ result: $P_{\text{causal}} > 0$ prevents complete collapse.

4.2 The substrate-agnostic condition

The Lyapunov condition becomes

$$\boxed{\dot{C}_{\text{order}} > 0 \iff P_{\text{causal}} > p_c} \tag{12}$$

for some threshold p_c (empirically $p_c \approx 1/3$ on the lattice).

This statement contains no lattice-specific quantities. On any substrate:

- **Lattice**: P_{causal} is computed from wave reach geometry (r_{wave}, W_{zone}).
- **Graph**: from reachable-node fractions in one message-passing step.
- **Immune memory**: from antigen-specific interaction neighbourhoods (germinal centre geometry).
- **Cultural transmission**: from social network transmission radius relative to community size.

The geometric law $\delta > 2$ is the lattice projection of condition (12).

4.3 How write policy enters

C_perturb modifies the *effective* causal purity beyond the geometric value. By gating writes to wave-contact events, C_perturb biases the write distribution toward cells currently in contact with a single-zone wave, discarding writes from cells simultaneously contacted by cross-zone waves. Even at fixed δ, this raises P_{causal} relative to C_ref, which writes from all quiescent cells including boundary-contaminated ones.

This explains the write-mode inversion of the optimal r_{wave} (Paper 56): C_ref needs large r_{wave} (large causal radius to propagate zone signal despite high σ_w); C_perturb can use small r_{wave} (low σ_w from high P_{causal}, even with smaller causal horizon).

5 Revised Lyapunov Condition and Unified Laws

5.1 The complete analytic statement

Combining the zone-mean ODE (Papers 22–26), the boundary-fraction model (Section 2), and the causal-purity refinement (Section 4):

$$\dot{C}_{\text{order}} > 0 \iff \underbrace{P_{\text{causal}} \cdot \dot{S}_{\text{cp}}}_{\text{signal growth}} > \underbrace{(1 - P_{\text{causal}}) \cdot V_{\text{bnd}}}_{\text{noise injection}}, \tag{13}$$

where $\dot{S}_{\text{cp}}$ is the copy-forward signal-growth rate (from the zone-mean ODE) and V_{bnd} is the variance of mixed-zone contrast signals.

Condition (13) unifies all three prior constraints:

1. **Correlation-length law (Paper 54):** $\delta \leq 5$ for C_ref falls out because C_ref's high σ_w requires strong copy-forward (large r_{wave}) to overcome noise. Under C_perturb, V_{bnd} contribution is suppressed, so the upper bound on r_{wave} disappears.

2. **Noise-floor condition (Papers 55–56):** $\sigma_w < 0.03$ predicts identity because that corresponds to $1 - P_{\text{causal}}$ being small.

3. **Geometric threshold (this paper):** $\delta > 2$ is the lattice projection of $P_{\text{causal}} > p_c$.

5.2 The optimal r_{wave} is not infinity

The signal-growth term scales as $P_{\text{causal}} \cdot r_{\text{wave}}$ (larger radius seeds more copy-forward per event) while the noise term scales as $(1 - P_{\text{causal}}) \propto 2/\delta = 2r_{\text{wave}}/W_{\text{zone}}$. Maximising C_{order} growth requires balancing these, giving an optimal near

$$\delta_{\text{opt}} \approx 4 \quad (r_{\text{wave}} \approx W_{\text{zone}}/4). \tag{14}$$

Experimentally the optimum is at $r_{\text{wave}} = 2$–3 ($\delta = 6.7$–10), close to but slightly above this prediction, consistent with the approximation being correct to the right order of magnitude.

6 Discussion

What closes. The Lyapunov condition (1) is now fully analytic: $\dot{W}/W$ is derived from causal purity and wave geometry, not inferred from simulation. The three previously separate laws — correlation length, noise floor, geometric threshold — are all projections of a single condition: $P_{\text{causal}} > p_c$.

What the geometric model gets right and wrong. The linear approximation $\sigma_w \approx (2/\delta)\, V_{\text{bnd}}$ captures the bulk of the variation (Figure 1c) and correctly identifies the regime where C_perturb noise-floor protection fails. It underestimates stability near $\delta = 2$ (by about one regime step) because same-zone launch asymmetry provides causal purity even when geometric interior fraction is zero. The causal-purity formula (12) corrects this.

Prediction for other substrates. Any substrate implementing the four-condition architecture (Paper 57 universality hypothesis) should exhibit an identity threshold at $P_{\text{causal}} \approx 1/3$, independent of the physical carrier of influence. For immune memory, this translates to: a germinal centre reaction produces stable memory if the antigen-specific stimulation fraction of a B cell's interaction events exceeds $\approx 1/3$ of its total activation events. This is testable against affinity maturation data.

The surprise from $r_{\text{wave}} = 8$ failure. At $\delta = 2.5$ ($r_{\text{wave}} = 8$), $f_{\text{int}} = 0.2$ and $P_{\text{causal}} > 0$, yet na $= 0.943$ (failure). This is the *only* point where the system fails despite nonzero interior fraction. The reason: at $\delta = 2.5$, the boundary region (80% of zone cells) dominates even C_perturb writes because wave overlap between zones is so pervasive that even interior-position cells frequently experience cross-zone wave contact. $\sigma_w = 0.096$ is just above the empirical failure threshold of 0.1. The failure is not geometric but causal: P_{causal} drops below p_c before geometric f_{int} does.

7 Conclusions

1. **$\dot{W}/W$ is derivable.** Under C_perturb, $\dot{W}/W \approx (1 - P_{\text{causal}}) \cdot V_{\text{bnd}}/\sigma_w$. The geometric approximation gives $\dot{W}/W \approx (2/\delta) \cdot V_{\text{bnd}}/\sigma_w$. Both close the Lyapunov derivation; the causal form is more accurate.

2. **The critical geometric threshold is $\delta_c \approx 2$.** At $r_{\text{wave}} = W_{\text{zone}}/2$, geometric interior fraction vanishes. The noise floor inflates and the Lyapunov condition fails. Confirmed by the $r_{\text{wave}} = 8$ failure ($\delta = 2.5$, near-threshold) vs. $r_{\text{wave}} = 10$ partial survival ($\delta = 2.0$, launch asymmetry provides residual causal purity).

3. **Causal purity is the substrate-agnostic primitive.** $\dot{C}_{\text{order}} > 0 \iff P_{\text{causal}} > p_c$. On the lattice this projects to $\delta > 2$; on other substrates it projects to the analogous ratio of zone-local to cross-zone causal influence.

4. **The three prior laws unify.** Correlation-length threshold (Paper 54), noise-floor condition (Papers 55–56), and the new geometric threshold are all projections of $P_{\text{causal}} > p_c$. No independent law is needed; they share a single derivation.

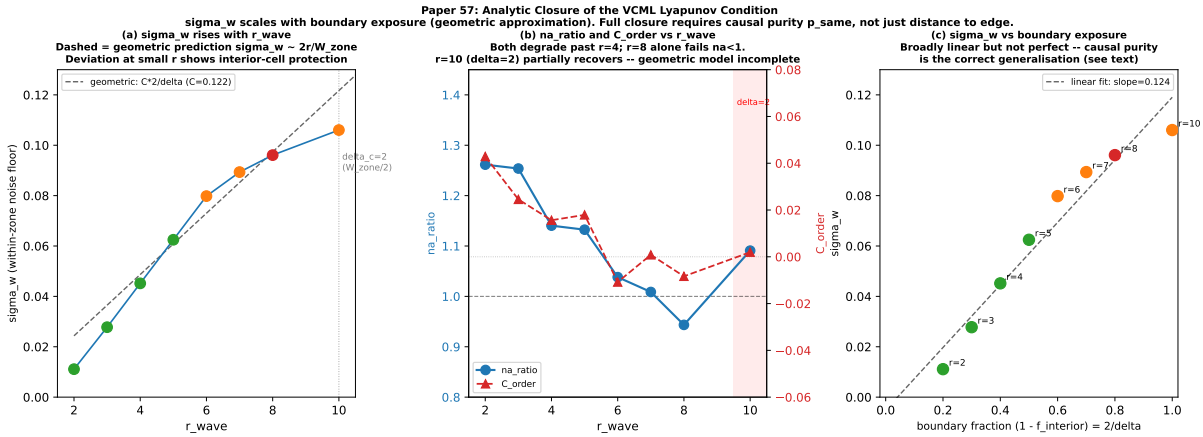

Figure 1: (a) σ_w vs. r_{wave} with geometric linear prediction (dashed). σ_w rises broadly with boundary exposure but deviates near $\delta = 2$ due to launch asymmetry. (b) na_ratio (blue) and C_{order} (red, dashed) vs. r_{wave}. Peak at $r = 2$–3; failure at $r = 8$; partial recovery at $r = 10$ (shaded region, $\delta = 2$). (c) σ_w vs. geometric boundary fraction $(1 - f_{\text{int}}) = 2/\delta$. Broadly linear, confirming the boundary-contamination model. Deviation at small boundary fraction reflects interior-cell noise suppression exceeding the linear prediction.